

Title

Implementation of Common Java Exceptions and Exception Handling

Objective

To understand and implement different types of Java exceptions using `try`, `catch`, and `throw` statements. The programs demonstrate common exceptions such as `ArrayIndexOutOfBoundsException`, `ArrayStoreException`, `ClassNotFoundException`, `IllegalArgumentException`, `InputMismatchException`, `IllegalStateException`, `NullPointerException`, `NoSuchMethodException`, `NumberFormatException`, and `StringIndexOutOfBoundsException`.

System Requirements

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Keyboard and Mouse

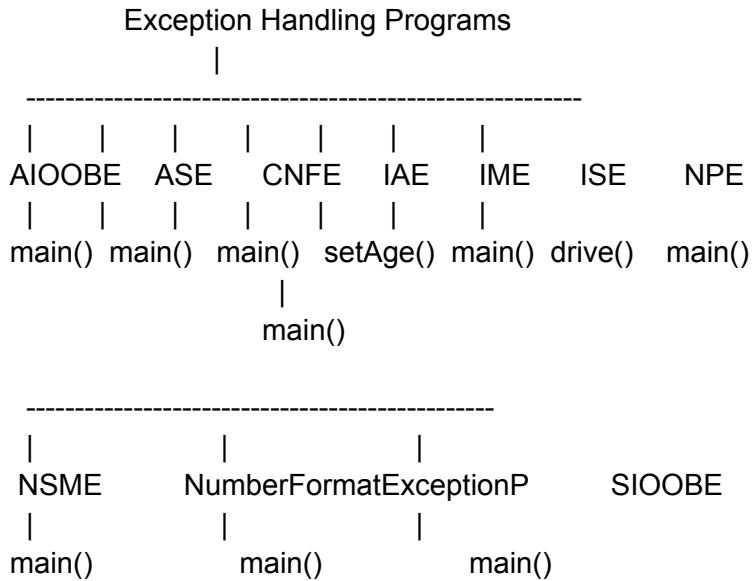
Software Requirements

- Windows/Linux/macOS
- Java JDK
- VS Code / Eclipse / IntelliJ IDEA / Command Prompt

Algorithm / Logic Description

1. Create separate Java programs for different exception types.
2. Take required input from the user wherever necessary.
3. Place statements that may generate exceptions inside the `try` block.
4. Use the appropriate `catch` block to handle the exception.
5. Display a suitable message when an exception occurs.
6. Use `throw` where an exception needs to be generated explicitly.
7. Compile and execute each Java program separately.
8. Observe the output for both handled and unhandled exceptions.

UML Class Diagram



Source Code

1. AIOOBE.java

```
import java.util.Scanner;

public class AIOOBE {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] arr = {10, 20, 30, 40, 50};

        System.out.print("Enter index: ");
        int index = sc.nextInt();

        try {
            System.out.println("Value = " + arr[index]);
        }
        catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Index is out of bounds");
        }

        System.out.println("Done");
        sc.close();
    }
}
```

2. ASE.java

```
public class ASE {  
    public static void main(String[] args) {  
  
        Object[] arr = new String[3];  
  
        arr[0] = "Hello";  
        arr[1] = "Java";  
  
        try {  
            arr[2] = 100;  
        }  
        catch (ArrayStoreException e) {  
            System.out.println("ArrayStoreException occurred");  
        }  
    }  
}
```

3. CNFE.java

```
class CNFE {  
    public static void main(String[] args) {  
  
        try {  
            Class.forName("Animal");  
            System.out.println("Animal class loaded");  
        }  
        catch (ClassNotFoundException e) {  
            System.out.println("Animal class not found");  
        }  
    }  
}
```

4. IAE.java

```
class IAE {  
  
    static void setAge(int age) {  
  
        if (age < 0) {  
            throw new IllegalArgumentException("Age cannot be negative");  
        }  
  
        System.out.println("Age is: " + age);  
    }  
}
```

```

    public static void main(String[] args) {

        setAge(-5);
    }
}

```

5. IME.java

```

import java.util.Scanner;
import java.util.InputMismatchException;

public class IME {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int a = 0;
        float b = 0;

        try {
            System.out.println("Enter int value for a");
            a = sc.nextInt();

            System.out.println("Enter a float value for b");
            b = sc.nextFloat();
        }
        catch (InputMismatchException e) {
            System.out.println("Enter proper values of a and b");
        }

        System.out.println("Entered values of a and b is: " + a + " " + b);

        sc.close();
    }
}

```

6. ISE.java

```

public class ISE {

    boolean started = false;

    void drive() {

```

```

        if (!started) {
            throw new IllegalStateException("Car is not started");
        }

        System.out.println("Car is driving");
    }

    public static void main(String[] args) {

        ISE c = new ISE();

        c.drive();
    }
}

```

7. NPE.java

```

class NPE {
    public static void main(String[] args) {

        String name = null;

        try {
            System.out.println(name.toUpperCase());
        }
        catch (NullPointerException e) {
            System.out.println("Cannot use a null object");
        }
    }
}

```

8. NSME.java

```

import java.lang.reflect.Method;

public class NSME {
    public static void main(String[] args) {

        try {
            Class<?> c = String.class;

            Method m = c.getMethod("hello");
        }
        catch (NoSuchMethodException e) {
            System.out.println("NoSuchMethodException occurred");
        }
    }
}

```

```

    }
}
}

```

9. NumberFormatExceptionP.java

```

import java.util.Scanner;

public class NumberFormatExceptionP {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter proper string to convert into integer");

        try {
            String s = sc.next();

            int num = Integer.parseInt(s);

            System.out.println("String is converted into int: " + num);
        }
        catch (NumberFormatException e) {
            System.out.println("Give proper string value to convert");
        }

        sc.close();
    }
}

```

10. SIOOBE.java

```

import java.util.InputMismatchException;
import java.util.Scanner;

public class SIOOBE {
    public static void main(String[] args) {

        String name = "naveen";

        Scanner sc = new Scanner(System.in);

        try {
            System.out.println("Enter value of i to print substring");

```

```

        int i = sc.nextInt();

        System.out.println(name.charAt(i));
        System.out.println(name.substring(i));
    }
    catch (StringIndexOutOfBoundsException e) {
        System.out.println("Index must be from 0 to 5");
    }
    catch (InputMismatchException e) {
        System.out.println("Enter i value only as integer");
    }
    }

    sc.close();
}
}

```

Compilation and Execution Commands

Each program is compiled and executed separately.

```

javac AIOOBE.java
java AIOOBE

```

```

javac ASE.java
java ASE

```

```

javac CNFE.java
java CNFE

```

```

javac IAE.java
java IAE

```

```

javac IME.java
java IME

```

```

javac ISE.java
java ISE

```

```

javac NPE.java
java NPE

```

```

javac NSME.java
java NSME

```

```
javac NumberFormatExceptionP.java
java NumberFormatExceptionP
```

```
javac SIOOBE.java
java SIOOBE
```

Output Screen

AIOOBE

```
Enter index: 7
Index is out of bounds
Done
```

ASE

```
ArrayStoreException occurred
```

CNFE

```
Animal class not found
```

IAE

```
Exception in thread "main" java.lang.IllegalArgumentException: Age cannot be negative
```

IME

```
Enter int value for a
10
Enter a float value for b
2.5
Entered values of a and b is: 10 2.5
```

ISE

```
Exception in thread "main" java.lang.IllegalStateException: Car is not started
```

NPE

```
Cannot use a null object
```

NSME

```
NoSuchMethodException occurred
```

NumberFormatExceptionP

```
Enter proper string to convert into integer
```

abc

Give proper string value to convert

SIOOBE

Enter value of i to print substring

10

Index must be from 0 to 5